U.S. Racial Stratification Through the Lens of Utility Burdens

An Indices of Informational Association Analysis of Complex Systems

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Abstract:

This article adopts joint informational association measures in piecewise order to analyze disparities in utility costs or the percentage of income a household spends on their water, electricity, and gas bills due to an individual's race, gender, and ethnicity in the US. Using Census Bureau data between 2008 and 2019, the research quantifies association measures for more than 12 million households. It finds that white men's heads of households consistently pay less for utilities (lower utility burdens) than white women, black men and women, non-white Hispanic men and women, and American Indian men and women. White men pay less for utilities if they are in the bottom 90% of the income distribution. This relative utility burden converges if black, non-white Hispanic, and American Indian households are in the top 10% of the income distribution. For this reason, the analysis uses joint piecewise information association measures to quantify the relative differences in the amount of information social scientists can infer about an individual household utility burden based on their social identity or the stratified grouping due to race, ethnicity, and sex. These measures verify that scientists can infer more informational value from households with social identities in racially stratified groups than their white counterparts. While providing a unique case study of informational association measures, this study advances the US reparations debate by analyzing the inherent racial disparities in consuming basic goods and not only on a household's income. Finally, these results illuminate how market-based economies create persistent utility burden inequalities.

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Introduction

What impact do race and gender have on what an individual pays for utilities (water, gas, electricity) in the United States? Commonly, in economics, a dominant method is evolutionary game theory to analyze how individuals maximize or satisfy their preferences, or strategies in the case of bounded rationality, within the constraints of cultural norms and values in certain areas. This paper takes a different approach by analyzing individuals within the complex systems they inhabit. To follow a Political Economy framework, researchers must select a method that addresses the complexities inherent in household utility payments. This study uses Shannon entropy as an alternative measure of inequality to evaluate disparities in utility burden income due to geography, race, and sex ([1]) [1]. Furthermore, the analysis examines how income inequality and informational associations impact the utility burden of individual households. The study introduces an alternative method of measuring income inequality and demonstrates systemic discrimination in utility payments based on social identity and relative income. It builds upon traditional methods of assessing inequalities in a capitalist system regarding natural resources and the consumption of basic goods. The argument examines discrimination in utility burdens through the lens of social identity and income decile.

The paper describes the use of Census Bureau data to construct the analysis. The third section, Method, explains the application of Shannon entropy and information association indices, comparing these to traditional measures of income inequality such as GINI and Theil. Section four analyzes the calculated metrics and figures that measure income and utility burden inequality. Section five evaluates the limitations and advantages of using information theory and Shannon entropy to measure inequality. Finally, section six suggests future directions for employing informational associations to extend the theoretical model and the policy implications to overcome racial discrimination in utility bills across the United States.

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1 Data

The sample is a survey from the Census Bureau Household Data (IPUMS) between 2008 and 2019. Out of the 51 million total households, it excludes farms, adjusted utility burden, and income for inflation, and it does not include imputed utility burdens (utility burden included in a household's rent). Also, the analysis adopts a method for adjusting the household size for income and utility burdens by adjusting the number of rooms in the household, suggested by dividing the PEW Research Center. Before diving into the data, we must point out our Annual utility burden measures. Then, the research places each individual, or head of household, in different group income deciles and Education attainment and combines race and sex to form an "Identity." After removing households in the top two percent of the income and utility burdens distribution and NAs created in the Identity variable, the total number of observations for the case study is 12,717,383. For further details, please find the attached R code in Appendices A and B.

2 Mutual Information and Its Probabilistic Content

2.1 Setting up the Inequality Measures

Investigating the likelihood that everyone pays the same utility burden regardless of identity or income decile requires a flexible, nonlinear association measure. Although annual water payments vary by usage, dwelling size, and season, we argue (following [**dosSantos2022**] and others) that water is a human right. To capture the social cost of water, we examine individual water payments Y and ask how much their distribution depends on identity or income decile X. To do so, we adopt mutual information (MI), interactions, and decompositions from [**dosSantos2022**].

Mutual information quantifies the reduction in uncertainty about Y when we learn X. Formally, let f_y be the marginal frequency distribution of Y and $f_{y|x}$ the conditional distribution given $X = x$. Then MI is defined as

$$I(Y;X) \;=\; H(Y) \;-\; H(Y \mid X) \;=\; H(Y) \;-\; \sum_x f_x \, H(Y \mid x)\,, \tag{1}$$

where $I(Y;X)$ measures in bits how much knowing X on average reduces uncertainty about Y.

Here,

$$H(Y) = -\sum_y f_y \log f_y \quad \text{and} \quad H(Y \mid x) = -\sum_y f_{y|x} \log f_{y|x}\,,$$

with each entropy term quantifying the heterogeneity of the corresponding distribution.

The minimum value $I(Y;X) = 0$ arises when Y and X are statistically independent, so observing X yields no new information about Y. Its maximum $I(Y;X) = H(Y)$ occurs when Y is a deterministic function of X, meaning $H(Y \mid X) = 0$.

2.1.1 A Non-Parametric Measure of Association

An equivalent symmetric form of MI is

$$I(Y;X) \;=\; H(Y) + H(X) - H(Y,X)\,, \tag{2}$$

where $H(Y,X)$ denotes the joint entropy of (Y,X), capturing the total uncertainty before observing either variable.

In terms of micro-configuration counts, one can write

$$I(Y;X) \;=\; \frac{1}{N}\,\log\frac{\mathbb{W}_{f_Y}\,\mathbb{W}_{f_X}}{\mathbb{W}_{f_{Y,X}}}\,, \tag{3}$$

which expresses MI as the log-ratio between the product of marginal configuration counts and the joint configuration count, normalized by sample size.

3 Information, Interactions, and Decompositions

3.1 Entropy as Degeneracy

Mutual information of a variable with itself reduces to its entropy:

$$I(Y;Y) = \frac{1}{N}\sum_i \log P\big(\sigma(y_i), f_Y, N, 1\big) \tag{4}$$

which counts, in logs, the number of micro-configurations consistent with the observed multiset of outcomes.

$$I(Y;Y) = H(Y) = -\sum_y f_y \log f_y = -\frac{1}{N}\sum_y n_y \log f_y = -\frac{1}{N}\log \pi(f_Y, f_Y, s, N)\,. \tag{5}$$

Thus the self-information equals the Shannon entropy of Y, itself a log-count of configurations.

3.2 The Conditional, Joint, and Total Information

The conditional mutual information between Y and Z given X is

$$I(Y;Z \mid X) = H(Y \mid X) - H(Y \mid X,Z) = \sum_x f_x \, I\big(Y;Z \mid x\big)\,. \tag{6}$$

This quantity measures how much knowing Z further reduces uncertainty about Y once X is already known.

$$I\big(Y;X,Z\big) = I(Y;X) + I(Y;Z \mid X)\,. \tag{7}$$

The chain-rule shows total information about Y from (X,Z) decomposes into marginal plus conditional contributions.

3.3 The Interaction Information

The three-way (interaction) information is

$$I(Y;X;Z) = I(Y;X) - I(Y;X \mid Z) = I(Y;X) + I(Y;Z) - I\big(Y;X,Z\big)\,, \tag{8}$$

which can be positive or negative, indicating synergistic or redundant interactions among the three variables.

$$I(Y;X;Z) = \frac{1}{N}\Big[\log \mathbb{P}\big(f_{Y|X,Z} \,\|\, f_{Y|Z}\big) - \log \mathbb{P}\big(f_{Y|X} \,\|\, f_Y\big)\Big]\,. \tag{9}$$

In this form, interaction information appears as a difference of log-likelihood ratios, quantifying how conditioning on Z alters the dependence between Y and X.

For a general set of covariates $\mathcal{V} = \{\mathcal{V}_1, \ldots, \mathcal{V}_k\}$, define incremental information

$$\mathcal{I}\big(Y;\mathcal{V}\big) \;\equiv\; I\big(Y;\mathcal{V}_1,\ldots,\mathcal{V}_{k-1}\big) \;-\; I\big(Y;\mathcal{V}_1,\ldots,\mathcal{V}_{k-1} \mid \mathcal{V}_k\big)\,, \tag{10}$$

which isolates the additional contribution of the kth covariate.

3.4 Decompositions of Information

One obtains the following decompositions:

$$I(Y) = I(Y;X) + I(Y \mid X) = I(Y;X) + \sum_x f_x \, I(Y \mid x), \tag{11}$$

splitting total information into between-group and within-group components.

$$I(Y;X) = I(Y;X;Z) + I(Y;X \mid Z) = I(Y;X;Z) + \sum_z f_z \, I(Y;X \mid z), \tag{12}$$

which further partitions pairwise information by a third variable.

$$\mathcal{I}(Y) = \mathcal{I}\big(Y \mid \mathcal{V}_1\big) + \mathcal{I}\big(Y;\mathcal{V}_1 \mid \mathcal{V}_2\big) + \cdots + \mathcal{I}\big(Y;\mathcal{V}_1,\ldots,\mathcal{V}_{k-1} \mid \mathcal{V}_k\big) + \mathcal{I}(Y;\mathcal{V}), \tag{13}$$

giving a full hierarchical decomposition of information contributions from a set of covariates.

4 Informational Association Measures on Utility Burdens: Empirical Evidence

4.1 Observable Economic Characteristics

Table 1: Informational Measures and Utility Burdens

Fig.	Characteristics	Indices	Main Results
1	Education	$\alpha(Y \parallel \chi_{educ})$	College education enhances variability in utility burdens, while individuals with lower educational attainment, particularly high-school dropouts, experience more consistent and predictable utility burdens.
2	Age	$\alpha(Y \parallel \chi_{age})$	The influence of age on utility burdens has become significantly more pronounced for individuals in their TBD.
3	Education, Age	$\alpha(Y \parallel \chi_{educ} \parallel \chi_{age})$	Higher education levels amplify diversity in returns from experience, with these returns peaking later in life for the more educated. Variations in returns across different educational groups continue to increase over time.
4	Gender	$I(X_{gender} \mid X_{educ})$	Across all educational achievement groups, being female strongly predicts utility burdens, while men experience greater diversity in utility burden differentiation.
5	Race/ethnicity, Gender	$I(X_{ident} \mid X_{educ})$, $I(X_{ident} \mid X_{age})$	Social identity consistently plays a significant role in predicting utility burdens for Black, American Indian, and Hispanic households. In contrast, White households show greater variation in utility burdens based on factors unrelated to social identity.
6	Race/ethnicity, Gender	$I(X_{ident} \mid X_{educ})$, $I(X_{ident} \mid X_{age})$	This observation aligns with observation 5 but provides additional detail on how gender and race contribute incrementally to differentiation in utility burdens.
7	Compound social identity	$I(X_{ident} \mid X_{educ})$, $I(X_{ident} \mid X_{age})$	Social identity significantly influences utility burdens for women from historically marginalized groups, whereas men from these groups have less variation compared to white men. Over time, the impact of social identity on utility burdens for white women has diminished.
8	Compound social identity and education	$m(Y; \chi_{id}; \chi_{ed})$	Informational synergies emerge between education, identity, and utility burdens for Black and white women with low levels of education, as well as for white individuals with high levels of education.
9	Compound social identity, education, relative utility burden	$m(Y; \chi_{id}; \chi_{ed}), S_{ed,id}$	As education and utility burdens increase, educational attainment becomes a weaker predictor of utility burdens for women. In contrast, white men exhibit the opposite trend, with educational attainment becoming a stronger predictor.

5 Discussion

The research reveals several significant and nuanced insights derived from the information measures, shedding light on how a person's social identity and income shape their relationship with water in a capitalist system. The figures demonstrate that utility burdens for white men show minimal informational variation, indicating limited predictability about what they pay for water. This suggests greater flexibility for white men in terms of water, electricity, and gas pricing. In stark contrast, utility burdens for American Indian, Black, and Hispanic women exhibit substantially higher informational variation, reflecting a stronger correlation between their social identities, income levels, and water costs. This disparity underscores the pronounced role of social identity and economic stratification in shaping utility burdens, offering deeper analytical perspectives on systemic inequities.

Conclusion

6 Appendix

6.1 A

Variable	**Code (Value)**	**Meaning**	**Reasoning**
FARM	1	Not a farm	We want to see individuals first, people who do not farm.
RELATE	1	Head of Household	Respondent is the head of the household.
SINK	2	Sink	Control for material conditions.
PLUMPING	20	With Complete Plumbing	Control for material conditions.
SHOWER	4	Has Shower	Control for material conditions.
HOTWATER	4	Hot and cold piped water	Control for material conditions.
FOODSTMP	1	Does not receive food stamps	Control for material conditions.
Inflation Measure		BLS.gov	(Actual Value/Index value)*100. Real not nominal Income, costs, wages.

Household Size	Pew Research Methodology	Adjusted income = Income / (number of people in HH)^n, where $n = 0.5$. Have to adjust data for number of people in the household.

6.2 B R Code for Analysis

Listing 1: R Code for Utility Burden Analysis

```
#### Packages ####
library(ipumsr)
library(dplyr, warn.conflicts = FALSE)
library(survey)
library(srvyr)
library(plyr)
library(dineq)
library(data.table)
library(tidyverse)
library(tibble)
library(reshape)
library(reshape2)
library(readr)
library(readxl)
library(ggplot2)
library(gridExtra)
library(scales)
library(knitr)
library(kableExtra)
library(frequency)
library(purrr)
library(stargazer)
library(sf)
library(stats)
library(spatstat.geom)
library(hrbrthemes)
```

```r
library(viridis)
library(formattable)

setwd("/Users/kurtsemm/Documents/Information_Theory/Utility_Burdens")
# Note that you can pass in the loaded DDI into the `read_ipums_micro()`
cps_ddi <- read_ipums_ddi("usa_00024.xml")
cps_data <- read_ipums_micro(cps_ddi, data_file = "usa_00024.dat.gz", verbose =
    FALSE)



## Clean data so water cost and income are greater than one.
## FARM equal to 1 means the Household is NOT a farm and we assume household  ##
     has complete plumbing. We leave household type alone for right now,
## also shower does not exist for the years we are looking at.
## Using constant PUMA instead of non-constant PUMA.

## Controlling for Material Conditions ##

cps_data%>% select(-SERIAL, -PERNUM)
cps_data <- cps_data %>%
  #select_if(is.labelled) %>%
  filter(COSTWATR >= 1,
         HHINCOME >= 1,
         FARM == 1,
         PLUMBING == 20,
         SINK == 2,
         SHOWER == 4,
         HOTWATER ==4,
         FOODSTMP == 1,
         RELATE == 1,
         !is.na(COUNTYFIP)) #COUNTYFIP goes back to 2005
#filter(SINK == 2) # We lose years 2001-2008, not included in survey before than
    .

```

```
library(readxl)
CPI <- read_excel("CPI.xlsx")
cpi <- CPI %>%
  select(YEAR, Annual)

#Combine cpi with our dataset & adjust for inflation
cps_data %>% group_by(YEAR)
cps_usa <- cps_data %>% left_join(cpi)

cps_usa <- cps_usa %>%
  mutate(HHINCOME = (HHINCOME/Annual)*100,
         COSTWATR = (COSTWATR/Annual)*100,
         COSTELEC = (COSTELEC/Annual)*100,
         COSTGAS =  (COSTGAS/Annual)*100)

# Next, we have to adjust income and water costs for number of people in
    household, number of rooms, number of bedrooms,
# Using https://www.pewresearch.org/social-trends/2011/10/03/appendix-b-
    adjusting-household-income-for-household-size/ as a reference.

n = 0.5

#Adjust for number of individuals in household.

cps_usa <- cps_usa %>%
  mutate(HHINCOME = (HHINCOME/(PERNUM^n)),
         COSTWATR = (COSTWATR/(PERNUM^n)),
         COSTELEC = (COSTELEC/(PERNUM^n)),
         COSTGAS =  (COSTGAS/(PERNUM^n)))

cps_usa<- cps_usa %>%
  mutate(
    util_burden = ((COSTWATR+COSTELEC+COSTGAS)/HHINCOME),
    elec_burden = (COSTELEC/HHINCOME),
    gas_burden = (COSTGAS/HHINCOME),
```

```
    water_burden = (COSTWATR/HHINCOME))
#### Create upper bound for histogram #####

#Utility Burden
util_pctile <- cps_usa %>%
  group_by(YEAR) %>%
  do(util_98 = reldist::wtd.quantile(.$util_burden, q = 0.98, weight = .$HHWT)) %>%
  mutate(util_98 = unlist(util_98)) %>%
  as.data.frame()

#Water Burden
water_pctile <- cps_usa %>%
  group_by(YEAR) %>%
  do(water_98 = reldist::wtd.quantile(.$water_burden, q = 0.98, weight = .$HHWT)) %>%
  mutate(water_98 = unlist(water_98)) %>%
  as.data.frame()

#Electricity Burden
elec_pctile <- cps_usa %>%
  group_by(YEAR) %>%
  do(elec_98 = reldist::wtd.quantile(.$elec_burden, q = 0.98, weight = .$HHWT)) %>%
  mutate(elec_98 = unlist(elec_98)) %>%
  as.data.frame()

#Gas Burden
gas_pctile <- cps_usa %>%
  group_by(YEAR) %>%
  do(gas_98 = reldist::wtd.quantile(.$gas_burden, q = 0.98, weight = .$HHWT)) %>%
  mutate(gas_98 = unlist(gas_98)) %>%
  as.data.frame()

```

```r


## Join columns pctile's to cps_usa and water_pctile.KS-8/14.
cps_usa <- cps_usa %>%
  left_join(util_pctile) %>%
  left_join(water_pctile) %>%
  left_join(elec_pctile) %>%
  left_join(gas_pctile)

util_list<-c("util_burden", "elec_burden", "gas_burden", "water_burden")

#### Head of Household social identity ####
# Age Level
cps_usa <- cps_usa %>%
  mutate(AGE = as.numeric(AGE)) %>%
  mutate(AGE_GRP = case_when(
    AGE >= 1 &  AGE <=  9 ~ "09 and Less",
    AGE >  9 &  AGE <= 19 ~ "19 and Less",
    AGE >  19 & AGE <= 29 ~ "29 and Less",
    AGE >  29 & AGE <= 39 ~ "39 and Less",
    AGE >  39 & AGE <= 49 ~ "49 and Less",
    AGE >  49 & AGE <= 59 ~ "59 and Less",
    AGE >  59 & AGE <= 69 ~ "69 and Less",
    AGE >  69 & AGE <= 79 ~ "79 and Less",
    AGE >  79 & AGE <= 89 ~ "89 and Less",
    AGE >= 89 ~ "89 & More",
    TRUE ~ NA_character_))

# Education Level
cps_usa <- cps_usa %>%
  mutate(EDUC = as_factor(EDUC),
         year = as.factor(YEAR)) %>%
  mutate(EDUC4 = factor(case_when(
    EDUC %in% c("N/A or no schooling", "Nursery school to grade 4",
                "Grade 5, 6, 7, or 8", "Grade 9", "Grade 10", "Grade 11")
```

```
    ~ "A: Less that HS",
    EDUC == "Grade 12" ~ "B: HS Finished",
    EDUC %in% c("1 year of college", "2 years of college", "3 years of college")
    ~ "C: Some College",
    EDUC %in% c("4 years of college", "5+ of college") ~ "D: College Grad",
    TRUE ~ NA_character_)))

# To create our Identity variable we need to create a new column using mutate()
# Identity by Race
cps_usa <- cps_usa %>%
  mutate(RACE = as_factor(RACE),
         HISPAN = as_factor(lbl_na_if(HISPAN, ~.val == 9)),
         SEX = as_factor(SEX)) %>%
  mutate(RACE_ETHD = case_when(
    HISPAN %in% c("Mexican", "Puerto Rican", "Cuban", "Other") ~ "hispanic",
    HISPAN == "Not Hispanic" & RACE == "White" ~ "white",
    HISPAN == "Not Hispanic" & RACE == "Black/African American" ~ "black",
    HISPAN == "Not Hispanic" & RACE == "Other Asian or Pacific Islander" ~ "asian",
    HISPAN == "Not Hispanic" & RACE == "American Indian or Alaska Native" ~ "american indian",
    HISPAN == "Not Hispanic" & RACE %in% c("Other race, nec", "non-Hispanic") ~ "other")) %>%
  mutate(RACE_ETH = factor(
    case_when(RACE_ETHD %in% c("other") ~ "other",
              RACE_ETHD %in% c("hispanic", "white", "black", "american indian", "asian") ~ RACE_ETHD)),
    RACE_ETHD = factor(RACE_ETHD)) %>%
  mutate(IDENTITY = factor(case_when(RACE_ETH == "white" & SEX == "Male" ~ "White Men",
                                     RACE_ETH == "white" & SEX == "Female" ~ "White Women",
                                     RACE_ETH == "black" & SEX == "Male" ~ "Black Men",
```

```
                                        RACE_ETH == "black" & SEX == "Female" ~ "Black Women",
                                        RACE_ETH == "hispanic" & SEX == "Male" ~ "Hispanic Men",
                                        RACE_ETH == "hispanic" & SEX == "Female" ~ "Hispanic Women",
                                        RACE_ETH == "american indian" & SEX == "Male" ~ "American Indian Men",
                                        RACE_ETH == "american indian" & SEX == "Female" ~ "American Indian Women",
                                        RACE_ETH == "asian" & SEX == "Male" ~ "Asian Men",
                                        RACE_ETH == "asian" & SEX == "Female" ~ "Asian Women",
                                        RACE_ETH == "other" & SEX == "Male" ~ "Other Men",
                                        RACE_ETH == "other" & SEX == "Female" ~ "Other Women",
                                        TRUE ~ NA_character_)))
summary(cps_usa$YEAR)
#2008-2019


sample<-cps_usa %>%
  mutate(util_pct = case_when(util_burden >= util_98 ~ 1, T ~ 0),
         water_pct = case_when(water_burden >= water_98 ~ 1, T ~ 0),
         elec_pct = case_when(elec_burden >= elec_98 ~ 1, T ~ 0),
         gas_pct = case_when(gas_burden >=gas_98 ~ 1, T ~ 0)) %>%
  filter(util_pct == 0,
         water_pct == 0,
         elec_pct == 0,
         gas_pct == 0)

summary(sample$util_burden)
saveRDS(sample, "sample_data.rds")
```

```r

for (i in 2008:2019){
  sample %>%
    filter(year==i) %>%
    ggplot(., aes(x=util_burden, fill=SEX)) +
    geom_histogram(color="#e9ecef", alpha=0.6, position = 'identity', binwidth =
        1) +
    scale_fill_manual(values=c("darkgreen", "blue")) +
    theme_ipsum() +
    labs(fill="")
  ggsave(histogram, file = paste(i,"util_burden.pdf"))
}

majority_util<-sample %>%
  filter(YEAR==2019)

ggplot(majority_util, aes(x = util_burden, fill = SEX)) +
  geom_density(alpha = 0.6) +
  scale_fill_manual(values = c("darkgreen", "blue")) +
  theme_minimal() +
  labs(x = "Util Burden", y = "Density", title = "Distribution of Util Burden")

#### Entropy Measure Functions ####
ent <- function(x){ ### Equation 1
  -sum(ifelse(x > 0, x * log(x), 0))
}

H_X <- function(D){
  ent(prop.table(apply(D, MARGIN = 1, FUN = sum)))
} ##Equation 3

H_z <- function(D){
  X_cond_Z = apply(D, MARGIN = c(1,3), FUN = sum)
  f_X_cond_Z = prop.table(X_cond_Z, margin=2)
  apply(f_X_cond_Z, MARGIN = 2, FUN = ent)
```

```
}

H_y <- function(D){ ### Equation 2
  X_cond_Y = apply(D, MARGIN = c(1,2), FUN = sum)
  f_X_cond_Y = prop.table(X_cond_Y, margin=2)
  apply(f_X_cond_Y, MARGIN = 2, FUN = ent)
}

H_z_y <- function(D){ ### Equation 5
  X_cond_Z_Y = prop.table(D, margin=c(2,3))
  apply(X_cond_Z_Y, MARGIN = c(2,3), FUN = ent)
}

histogrammer2 <- function(var, top, weight) {
  breaks_flex=seq.int(from=0, to=top, length.out=34)
  hist=spatstat.geom::whist(var, breaks=breaks_flex, weights=weight)
  return(data.frame(freq=hist,freq=breaks_flex[-1]-diff(breaks_flex)/2))
}
#### Create Histogrammer ####

freqr <- function(var, top, weight, k = 34) { # takes the variable to be binned, the upper limit, and the survey weight
  breaks_flex = seq.int(from=0, to=top, length.out = k)
  as.numeric(spatstat.geom::whist(var, breaks = breaks_flex, weights=weight)) # returns the weighted frequencies per bin
}

midr <- function(var, top, weight, k = 34) { # takes the variable to be binned, the upper limit, and the survey weight
  breaks_flex = seq.int(from=0, to=top, length.out = k)
  breaks_flex[-1]-diff(breaks_flex)/2 # returns the bin mid-points
}

Nr <- function(var, top, k = 34) { # takes the variable to be binned, the upper limit
```

```
  breaks_flex = seq.int(from=0, to=top, length.out = k)
  hist(var, breaks = breaks_flex, plot = FALSE)$counts # returns the sample size per bin
}

freqr_result <- freqr(var = sample$util_burden, top = sample$util_98,
                      weight = sample$HHWT)

set.seed(0308)
util_xyz_histogram <- sample %>%
  filter(!is.na(EDUC4),
         !is.na(IDENTITY)) %>%
  dplyr::group_by(year, IDENTITY, EDUC4, .drop=FALSE) %>%
  dplyr::reframe(
    freq = freqr(util_burden, util_98, HHWT),
    mid_burden = midr(util_burden, util_98, HHWT),
    N = Nr(util_burden, util_98)
  ) %>%
  mutate(rel_freq = freq/sum(freq)) %>%
  filter(freq > 0) %>%
  mutate(Sex=factor(case_when(
    IDENTITY %in% c("Black Men",
                    "Hispanic Men",
                    "Asian Men",
                    "American Indian Men",
                    "Other Men",
                    "White Men")~"Men",
    IDENTITY %in% c("Black Women",
                    "Hispanic Women",
                    "Asian Women",
                    "American Indian Women",
                    "Other Women",
                    "White Women") ~ "Women")),
    Ethnicity=factor(case_when(
      IDENTITY %in% c("Black Men","Black Women") ~ "Black",
```

```
      IDENTITY %in% c("Hispanic Men","Hispanic Women") ~ "Hispanic",
      IDENTITY %in% c("Asian Men","Asian Women") ~ "Asian",
      IDENTITY %in% c("American Indian Men","American Indian Women") ~
        "American Indian",
      IDENTITY %in% c("Other Men","Other Women") ~ "Other",
      IDENTITY %in% c("White Men","White Women") ~ "White")))

#### Create Information Measures ####
A_X_z <- function(d, x, y, z){ # Informational association of Z and X, Equation
    6
  form <- deparse(substitute(x~y~z))
  D = reshape2::acast(d, formula = form, value.var = "freq",
                      fun.aggregate = length)
  ret = 1 - H_z(D)/H_X(D)
  ret2 = data.frame(A_X_z = ret, row.names = NULL)
  ret2$temp = as.character(names(ret))
  names(ret2)[2] <- deparse(substitute(z))
  return(ret2)
}

A_X_y <- function(d, x, y, z){  # Informational association of Y and X Equation
    6
  form <- deparse(substitute(x~y~z))
  D = reshape2::acast(d, formula = form, value.var = "freq",
                      fun.aggregate = length)
  ret = 1 - H_y(D)/H_X(D)
  ret2 = data.frame(A_X_y = ret, row.names = NULL)
  ret2$temp = as.character(names(ret))
  names(ret2)[2] <- deparse(substitute(y))
  return(ret2)
}


I_z_y <-  function(d, x, y, z){# Incremental info assoc of Z, X when Y is known
  form <- deparse(substitute(x~y~z))
```

```
  D = reshape2::acast(d, formula = form, value.var = "freq",
                      fun.aggregate = length)
  ret = (H_y(D)-H_z_y(D))/H_X(D)
  ret2 = data.frame(ret, row.names = NULL)
  ret2$temp = as.character(rownames(ret))
  ret3 = gather(ret2, key = temp2, value = "I_z_y", -temp)
  names(ret3)[1] <- deparse(substitute(y));
  names(ret3)[2] <- deparse(substitute(z))
  return(ret3)
} ### Equation

I_y_z <-  function(d, x, y, z){# Incremental info assoc of Y, X when Z is known
  form <- deparse(substitute(x~y~z))
  D = reshape2::acast(d, formula = form, value.var = "freq",
                      fun.aggregate = length)
  ret = t((H_z(D)-t(H_z_y(D)))/H_X(D))
  ret2 = data.frame(ret, row.names = NULL)
  ret2$temp = as.character(rownames(ret))
  ret3 = gather(ret2, key = temp2, value = "I_y_z", -temp)
  names(ret3)[1] <- deparse(substitute(y));
  names(ret3)[2] <- deparse(substitute(z))
  return(ret3)
}

M_x_y_z <-  function(d, x, y, z){ # Interaction information of Y, Z, for X
  form <- deparse(substitute(x~y~z))
  D = reshape2::acast(d, formula = form, value.var = "freq",
                      fun.aggregate = length)
  ret = ret = (H_X(D) - H_y(D) - t(H_z(D)-t(H_z_y(D))) ) / H_X(D)
  ret2 = data.frame(ret, row.names = NULL)
  ret2$temp = as.character(rownames(ret))
  ret3 = gather(ret2, key = temp2, value = "M_x_y_z", -temp)
  names(ret3)[1] <- deparse(substitute(y));
  names(ret3)[2] <- deparse(substitute(z))
  return(ret3)
```

```
}
```

References

[1] Paulo L. dos Santos and Noé Wiener. "Indices of Informational Association and Analysis of Complex Socio-Economic Systems". In: *Entropy* 21.4 (2019). ISSN: 1099-4300. DOI: 10.3390/e21040367. URL: https://www.mdpi.com/1099-4300/21/4/367.